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Continuous Assessment Test (CAT) – I – July 2026

Programme	:	**B.Tech**	Semester	:	**Fall Semester 2026-27**
Course Title	:	**Computer Architecture and Organization**	Course Code	:	**CSE2003**
Date/Session	:	**10-8-2026 / S-III**	Slot	:	**C14+E11+E12**
Time	:	**1 ½ hours**	Max. Marks	:	**50**

Answer all the Questions

Q.No.	Sub. Sec.	Question Description	Marks
1		Identify different components of the instruction for given instruction: LOAD R1, LOCA Where R1 is a processor register and LOCA represents memory address. State the lifecycle of the above instruction's execution.	**10**
2		Define performance measures-MIPS and Execution Time. Two different compilers are being tested for a 500 MHz machine with three different classes of instructions: Class A, Class B, and Class C, which require two, four, and three cycles (respectively). Both compilers are used to produce code for a large piece of software. The first compiler's code uses 7 million Class A instructions, 2 million Class B instructions, and 3 million Class C instructions. The second compiler's code uses 10 million Class A instructions, 2 million Class B instructions, and 1 million Class C instructions. (i)Which sequence will be faster according to MIPS? (ii)Which sequence will be faster according to execution time?	**10**
3		i).Find using two's complement arithmetic: [14 -37] (ii)Consider the two 8-bit numbers A =01000010 and B = 10000100. Add the two binary numbers and interpret the sum assuming that the numbers are (1) unsigned, and (2) signed.	**10**
4		Use Non-restoring division algorithm to compute the division taking dividend as 12 and divisor as 5. Illustrate clearly the complete step by step procedure to compute the results.	**10**
5		Write a assembly language program to evaluate the given arithmetic statement: $$Z = \frac{[A * (B + C)] - D]}{(E + F - G)}$$ a) Using an accumulator type computer with one-operand instructions. b) Using a general register load store computer with two-operand instructions. c) Using a stack organized computer with zero address instructions (stack instructions) Do not modify the values of A, B, C, D, E, F or G. Use a temporary location T to store the intermediate results if is necessary.	**10**

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